

R3 v4

AI-Native Digital Audio Workstation

Product Requirements Document — Version 4.0

Version	4.0 — Engineering-Verified Investor-Grade
Status	Active — MVP Build Phase (Items 1–3 Complete)
Updated	2026 Q2
Classification	Confidential
Audience	Engineering, Investors, Design, QA
Owner	Ty (Founder / Lead Engineer)
Codebase	~/Stable — pnpm monorepo — feature/lpte-extraction
Deployment	Railway (backend) + Vercel (frontend)
Hygiene Baseline	TSC: 0 errors — 11 routers wired — aiDecisionLog migration 0005

CHANGELOG FROM PRD v3: LemonSqueezy removed. Tier names corrected to explorer/creator/pro_artist. State management corrected from Redux to Zustand. appRouter shape updated to reflect 11 wired routers. aiDecisionLog table added. SessionChip + SessionSummaryPanel wired. Future Development section added.

0. Executive Summary

R3 v4 is an AI-native Digital Audio Workstation built for the market existing tools have failed: DJs and music creators who need professional-quality results without professional-level technical skill or time investment.

The global music production software market is valued at \$1.4B (2024) and growing at 8.2% CAGR. No dominant player has shipped a genuinely AI-native workflow layer. Ableton, Logic, and Rekordbox remain fundamentally unchanged in their core interaction model for over a decade.

R3's differentiation is architectural, not cosmetic. The Low-Latency Processing Transition Engine (LLPTE) is a named, inspectable, real-time AI pipeline with confirmed sub-15ms inference latency. Every AI decision is transparent, overridable, and quantifiably faster than its manual equivalent.

Current Build State (Verified 2026-04-09)

MVP Item	Status	Evidence
AI Auto-Leveling	✔ Complete	6 layers, 20 Vitest tests, LLPTE wired
Smart Transitions	✔ Complete	9 files, 22 Vitest tests, Camelot scoring
Time Savings Tracking	✔ Wired	SessionChip + SessionSummaryPanel in DAW.tsx
Mix Suggestion System	❑ In Progress	MixSuggestionsPanel.tsx built, backend pending
aiDecisionLog table	✔ Migration 0005	11 columns, Drizzle schema + types
appRouter	✔ 11 routers	sessions, sessionMetrics, admin, daw, subscription, mixer, dj, aiMix, projects, presets, settings
TSC	✔ Zero errors	Verified post all patches
Railway deployment	✔ Live	railway.toml confirmed

One-sentence pitch: R3 makes professional-quality mixing accessible to anyone with a track and a deadline — and proves it with a number on screen.

1. Product Summary & Vision

Identity

Product Name	R3 v4
Category	AI-Native Digital Audio Workstation (DAW)
Build Target	TypeScript pnpm Monorepo
Frontend	React 18 + Vite 5 (client/)
Backend	Express + tRPC + WebSocket (server/)
Database	PostgreSQL via Drizzle ORM 0.39.3
AI Pipeline	LLPTE — 6 monorepo packages (packages/llpte-*)
Rendering	WebGPU + Three.js r128 / React Three Fiber
Auth	bcrypt v6.0.0 + JWT (jsonwebtoken v9.0.3)
Payments	Stripe v20.4.1 ONLY — no LemonSqueezy
State Management	Zustand (NOT Redux — corrected from PRD v3)
Routing	Wouter (NOT react-router-dom)
Testing	Vitest — 42+ cases across LLPTE layers
Deployment	Railway (backend) + Vercel (frontend)
Design System	Tailwind CSS + Radix UI
Tiers	explorer · creator · pro_artist (Stripe only)

Vision Statement

The future of music creation is not a smarter manual interface. It is a system that understands audio as well as the engineer sitting next to you — one that can hear what you cannot, adjust what you would have missed, and do it before you knew it was a problem.

Design Philosophy

- Dark by default. Every background is zinc-950 or darker.
- Thin borders, maximum information. 1px borders at 6% opacity. Zero decorative chrome.
- Neon as signal, not decoration. Cyan = active. Violet = AI. Amber = warning. Red = danger.
- JetBrains Mono for data, Inter for language. BPM, latency, dB — monospace. Labels — Inter.
- Every AI action leaves a visible trace. Ghost knobs, violet overlays, confidence scores.

2. Market Opportunity

Total Addressable Market

Segment	Size	Source
Global Music Production Software	\$1.4B (2024)	Grand View Research
CAGR through 2030	8.2%	Grand View Research
Projected Market (2030)	\$2.3B	Calculated
DJ Software Segment	\$380M	Statista
Content Creator Audio Tools	\$620M	IBISWorld

Serviceable Addressable Market

- Digital DJs: 4.2M active globally (DJPA 2023). 1.8M use software-based tools. Average spend: \$180/year. SAM: \$324M.
- Content Creators: 50M+ active (YouTube, TikTok, Podcasts). 12M spend on audio tools. Average spend: \$85/year. SAM: \$1.02B.
- Combined SAM: approximately \$1.34B.

Why Now

- WebGPU is production-ready. Real-time GPU-accelerated audio visualization in a browser-native desktop application — no plugin required.
- Quantized inference is fast enough. Sub-15ms inference on consumer hardware was not achievable 24 months ago. R3 ships at 10ms.
- The creator economy demands professional output. Audiences trained by Spotify and Apple Music reject amateur mixes. R3 solves this.

3. Competitive Landscape

Direct Competitors

Dimension	Ableton Live 12	Traktor Pro 4	R3 v4
AI Integration	None (Max for Live only)	None	Native LLPTE pipeline
Learning Curve	6–12 months	3–6 months	Target: 30 minutes
Price	\$99–\$749 one-time	\$99 one-time	\$20/month (SaaS)
AI Transparency	N/A	N/A	Ghost knobs + confidence scores
Time Savings	None	None	Native — session-level tracking
Platform	Desktop only	Desktop only	Desktop-first, web-ready

Competitive Moat

R3's moat is not the AI model. The moat is the LLPTE architecture: a named, versioned, inspectable real-time pipeline embedded in the product's visual identity. Competitors cannot copy the node graph, the ghost knobs, or the Time Savings panel without rebuilding their entire product architecture.

Secondary moat — data flywheel: Every accepted AI suggestion is a labeled training signal. At 5,000 users with 3 sessions/week, R3 accumulates approximately 780,000 labeled audio engineering decisions per month. This dataset has no competitor equivalent.

4. Business Model & Monetization

Subscription Tiers (Stripe Only — Corrected from PRD v3)

Note: PRD v3 incorrectly listed tier names as Starter/Pro/Studio. Correct names in codebase and Stripe: explorer / creator / pro_artist.

Tier (DB value)	Price	Target User
explorer	Free	First-time creators, conversion target
creator	\$20/month	Primary commercial tier — DJs and serious creators
pro_artist	\$60/month (Post-MVP)	Professional producers and small studios
Enterprise / API	Custom Pricing (Post-MVP)	Platforms, labels, hardware manufacturers

Revenue Projections (Conservative)

Quarter	Paying Users	MRR	ARR
Q1 Post-Launch	500	\$10,000	\$120,000
Q2	1,500	\$30,000	\$360,000
Q3	3,500	\$70,000	\$840,000
Q4	6,000	\$120,000	\$1,440,000
Year 2 Q4	25,000	\$500,000	\$6,000,000

Unit Economics

CAC (content-led)	≤\$35
LTV at \$20/month	\$240 (12-month avg retention)
LTV:CAC Ratio	≥7:1
Gross Margin	82%
Monthly Churn Target	≤4%
Payback Period	≤2 months

5. Target Users — Deep Personas

Persona 1: Marcus — The Working DJ

Age / Location	28, Chicago
Income	\$200–\$500/night, 3–4 nights/week
Tools Today	Rekordbox + Pioneer CDJ + Ableton for prep
Core Pain	Set prep takes 8 hours. Should take 4.
Willingness to Pay	\$20/month without hesitation if transitions are provably better
Acquisition Channel	r/DJs, DJTT, short-form video

Persona 2: Priya — The Content Creator

Age / Location	24, Austin
Audience	180K YouTube subscribers, lifestyle/travel
Audio Skill	Near zero — treats audio as afterthought
Core Pain	Gets comments about audio quality. Cannot fix it.
Willingness to Pay	\$20/month if it saves 1 hour/week (\$100+ hourly value)
Acquisition Channel	Creator economy newsletters, "pro audio in 10 min" content

Persona 3: Deon — The Bedroom Producer

Age / Location	19, Atlanta
Platform	FL Studio, SoundCloud, seeking A&R placements
Audio Skill	High for sound design, low for mixing/mastering
Core Pain	Mixes are "muddy." Knows it. Cannot fix it.
Willingness to Pay	\$20/month if it gets him a placement or serious mix compliment
Acquisition Channel	SoundCloud community, beat-making subreddits, TikTok

6. Objectives & Success Criteria

Strategic Objectives (Ranked)

- Demonstrate that AI produces a measurably better mix than a human in less time — proven by the Time Savings panel and before/after audio comparison.
- Establish LLPTE as a named, defensible, visible AI architecture that differentiates R3 from every competitor.
- Achieve 65+ NPS among alpha users within 60 days of private launch.
- Reach \$120K ARR within 12 months of public launch.
- Collect 1M+ labeled AI decision signals for model improvement within 18 months.

Feature-Level Success Criteria

Feature	Metric	Target	Status
AI Auto-Leveling	Acceptance rate	≥65%	✔ Built — tracking via aiDecisionLog
AI Auto-Leveling	Inference latency	≤15ms	✔ 10ms confirmed
Smart Transitions	Setup time reduction	≥50%	✔ Built
Time Savings	Panel adoption	≥80% of sessions	✔ Wired in DAW.tsx
Mix Suggestion	Acceptance rate	≥40%	❑ MVP item 4 pending
Overall	Day 7 retention	≥60%	❑ Pre-launch
Overall	Day 30 retention	≥40%	❑ Pre-launch

7. UI Architecture — Five Zones (Verified)

Zone 1 — Top Navigation Bar (48px, full-width)

- Background: absolute black. Zero transparency.
- Left: R3 wordmark — JetBrains Mono, weight 700, letter-spacing 0.2em, pure white.
- Center: Transport controls — skip-back, stop, record, play (cyan glow #00F5FF when active), skip-forward.
- Center: BPM counter — JetBrains Mono, 2 decimal places, tap-tempo adjacent.
- Right: CPU/memory meter, user avatar, tier badge (Pro = violet pill), settings gear.
- Right: SessionChip — shows "● Live" in violet during active session, AI action count. (WIRED 2026-04-09)

Zone 2 — Left Sidebar — Project & Sample Browser (240px)

- Background: zinc-900. Right border: 1px zinc-800.
- Search field — debounced 200ms — queries tRPC sampleSearch procedure.
- File tree: PROJECTS, SAMPLES, MIDI, PRESETS, UPLOADS — chevron collapse/expand.
- Mini waveform preview on hover — 48px height, electric blue #3B82F6.

Zone 3 — Center Arrangement / Timeline (~60% width)

- Background: zinc-950. Playhead: 1px solid #00F5FF with triangle handle and 4px radial feather.
- Eight tracks: KICK, 808 BASS, SYNTH LEAD, CHORD PAD, VOCAL CHOP, FX SEND, AI MIX (LLPTE overlay), MASTER.
- Right-click context menu on clips: Rename, Duplicate, Delete, Bounce to Audio, Send to AI.
- "Send to AI" triggers LLPTE pipeline — violet overlay renders — transition selector appears.

Zone 4A — DJ Mixer / Channel Strips (bottom-left, 220px tall)

- Six vertical channel strips with 3-band EQ, gain trim knob, VU meter, fader.
- Ghost knobs appear when AI Auto-Leveling is active — translucent violet overlay.
- DJ crossfader with AI-suggested position indicator during transitions.

Zone 4B — LLPTE Node Graph (bottom-right)

- Five animated nodes: inputRouter → spectralAnalyzer → llpte-core → transitionGraph → outputBus.
- Connector lines: 1px animated dashes, 5px dash / 5px gap, 30px/second signal direction.
- LLPTE Core: 2× size, arc spinner, inference 10ms badge, pulsing ring during inference.
- Transition Graph tooltip: "847 active edges | last tick 0.8ms" — 200ms hover delay.
- Signal oscilloscope: violet (pre-AI) and cyan (post-AI), 30fps, 40px height.

Zone 5 — Right Sidebar — Effects Rack (280px)

- Six modules: Reverb, Delay, Compressor, Transition Graph (LLPTE), Output Bus (LLPTE), EQ.
- Each module: power toggle, drag handle, expand arrow, active indicator (emerald dot).

8. Core Features — Full Specification

8.1 AI Auto-Leveling — Complete

Gain staging is the most common source of amateur-sounding mixes. R3 addresses it automatically within the first 2 bars of playback via the LLPTE pipeline.

Confidence Gating Logic

- confidence $\geq$ 0.65 → auto-apply via `AudioParam.setTargetAtTime()`
- confidence $\geq$ 0.40 → surface as ghost knob suggestion only
- confidence $<$ 0.40 → discard silently, log to `aiDecisionLog`

8.2 Smart Transitions — Complete

Seven transition types precomputed by `transitionGraph` before the phrase boundary arrives.

Type	Mechanism	Best For
Crossfade (linear)	Equal-power gain crossfade	Same-key tracks
Crossfade (S-curve)	Sigmoidal gain curve	Energy preservation
Filter Sweep	LPF close → HPF open on incoming	High-energy drops
Reverb Tail	Outgoing reverb extended, incoming fades in	Atmospheric sections
Beat Drop Alignment	Transition delayed to nearest downbeat	Drop-centric sets
Echo Freeze	Outgoing pitch-locked, incoming crossfades	Dramatic transitions
Key-Match Crossfade	Camelot wheel harmonic scoring	Maximum harmonic quality

8.3 Time Savings Tracking — Wired (SessionChip + SessionSummaryPanel)

Every session tracked against manual baseline. Displayed in top nav chip and full-screen summary overlay on session end.

Action Type	Manual Baseline	AI Execution
Gain adjustment (one track)	45 seconds	0.5 seconds
EQ sweep (one track)	90 seconds	1 second
Crossfade transition setup	180 seconds	5 seconds
Frequency conflict detection	300 seconds	0.1 seconds

8.4 Mix Suggestion System — In Progress (MVP Item 4)

MixSuggestionsPanel.tsx component exists. Trigger detection logic in server/services/. Backend wiring to sessionMetrics.router pending.

- Frequency masking detection: FFT overlap in shared bands > -12dBFS for ≥1 bar.
- Dynamic imbalance: RMS differential > 6dB between any two active tracks.
- Arrangement conflict: ≥3 tracks simultaneously peaking above -6dBFS.

8.5 LLPTE Core — Real-Time AI Engine

Package Architecture

packages/llpte-core/	– Pipeline orchestration, lifecycle
packages/llpte-adapters/	– inputRouter, format normalization
packages/llpte-signal/	– spectralAnalyzer, FFT/RMS/LUFS
packages/llpte-ai/	– aiMixEngine, inference, confidence
packages/llpte-transition-graph/	– transitionGraph, curve precompute
packages/llpte-execution/	– outputBus, signal delivery

Performance Contract (SLA)

Metric	Hard Limit	Current (Verified)
Inference latency (p50)	≤15ms	10ms
Inference latency (p99)	≤25ms	TBD
Node tick time	≤1ms	0.8ms
Active edges (MVP)	≤2000	847
Pipeline uptime	99.9% per session	Auto-restart on breach

9. User Flows

Primary Flow: DJ Set Preparation

- Entry: User opens R3, lands on arrangement view (route: /instrument → /daw).
- Step 1 — Track load: drag tracks from sidebar, LLPTE pipeline initializes.
- Step 2 — First playback: Space bar, playhead starts, LLPTE activates within 2 bars.
- Step 3 — Auto-leveling: Toast fires "AI Mix updated — confidence 94%", ghost knobs appear.
- Step 4 — Frequency conflict: Suggestion panel fires, user accepts EQ adjustment.
- Step 5 — Transition zone: Violet overlay appears at bar 15, user selects Key-Match Crossfade.
- Step 6 — Session end: Space bar, SessionSummaryPanel appears, user exports PNG.

Routing Contract (Non-Negotiable)

Post-login redirect	/instrument ONLY — never /daw
User journey	Pricing → Login → Instrument → DAW → Loopstation
ProtectedRoute	Must NOT call hydrateFromToken() during render

10. Non-Goals (Firm Scope Boundary)

Non-Goal	Why Excluded	When Revisited
VST/AU/AAX plugin support	Requires native binary bridge — months of work	Post-Series A
Advanced MIDI composition	Separate product surface	Roadmap Phase 3
Real-time collaboration	Requires conflict resolution logic	Roadmap Phase 2
Mobile platform	Requires AudioWorklet + SharedArrayBuffer (desktop APIs)	Post \$5M ARR
Sound design / synthesis	Different user and use case	Never (separate product)
Offline mode	Cloud inference preferred for model updates	Roadmap Phase 2

11. Technical Requirements & Architecture

Performance Targets

Layer	Metric	Requirement
Audio engine	Round-trip latency	≤10ms
LLPTE inference	p50 latency	≤15ms (current: 10ms)
LLPTE inference	p99 latency	≤25ms
Node tick	Per-node	≤1ms (current: 0.8ms)
UI frame rate	During active playback	≥60fps
WebSocket round-trip	State sync	≤50ms
Database query	Session write	≤100ms
Memory ceiling	8 tracks × 5 min	≤2GB RAM

Stack (Pinned Versions — Verified)

TypeScript	5.9.3
drizzle-orm	0.39.3
ws	8.20.0
Three.js	0.128.0 (r128)
Stripe	20.4.1
Express	4.22.1
Zod	3.25.76
Vitest	Latest — coverage via v8
pnpm	10.33.0
Node	22.x (Active LTS — system confirmed v22.22.0)

HARD GUARDS — Non-Negotiable (CLAUDE.md)

- No any — use unknown + type guard. (15 violations fixed 2026-04-09. 5 remaining: routes/presets.ts × 4, audio-analysis.ts × 1 with eslint-disable exemption — no .d.ts for node-web-audio-api)
- No swallowed exceptions — all async functions handle errors explicitly.
- No console.log in committed code. (5 violations in server/index.ts:300-308 — pending fix)
- No write without read first (Wire.txt protocol).
- No LemonSqueezy tier strings — ever.
- Post-login redirect: /instrument only, never /daw.
- No hydrateFromToken() inside ProtectedRoute render.

12. Data Architecture & Schema

PostgreSQL Tables (Verified — server/db/schema.ts)

Table	Key Columns	Status
users	id, email, hashedPassword, tier, isAdmin, createdAt	✔ Live
sessions	id, userId, startedAt, endedAt, metrics JSON	✔ Live
sessionMetrics	id, sessionId, aiActionsCount, timeSavedMs, acceptanceRate	✔ Live
subscriptions	id, userId, tier, stripeCustomerId, status	✔ Live
projects	id, userId, name, arrangementJSON, createdAt	✔ Live
samples	id, userId, filename, path, duration, sampleRate, bitDepth	✔ Live
presets	id, userId, name, type, data, isFactory	✔ Live
settings	id, userId, 17 columns, updatedAt	✔ Live
aiDecisionLog	id, sessionId, nodeId, actionType, trackId, inputConfidence, displayedConfidence, decision, outcome, latencyMs, timestamp	✔ Migration 0005 (2026-04-09)
effectPresetsTable	id, name, settings, createdAt, updatedAt	✔ Live
effectChainsTable	id, name, nodes, createdAt, updatedAt	✔ Live
djCuesTable	id, trackId, position, label, color, createdAt	✔ Live
waveformEditsTable	id, sampleId, editType, params, createdAt	✔ Live

AIDecisionLog — Key Schema (PRD §12 — Now Implemented)

```
id:                text PRIMARY KEY
sessionId:         text NOT NULL (FK to sessions)
nodeId:           text NOT NULL -- 'aiMixEngine' | 'transitionGraph'
actionType:       text NOT NULL -- 'gain_adjust' | 'eq_suggest' |
'transition_generate' | 'conflict_flag'
trackId:          text NULLABLE
inputConfidence:  real NOT NULL -- raw model confidence [0,1]
displayedConfidence: real NOT NULL -- after calibration
decision:         jsonb NOT NULL -- action-specific payload
outcome:          text NOT NULL -- 'auto_applied' | 'accepted' | 'rejected'
| 'ignored' | 'discarded'
latencyMs:        integer NOT NULL
timestamp:        text NOT NULL
```


13. API Contract Specification

tRPC appRouter Shape (Verified — server/procedures.ts, 2026-04-09)

Router Key	File	Status
sessions	routers/sessions.ts	✔ Wired
sessionMetrics	routers/sessionMetrics.router.ts	✔ Wired
admin	routers/adminRouter.ts	✔ Wired
daw	routers/daw.ts	✔ Wired
subscription	routers/subscription.ts	✔ Wired
mixer	routers/mixer.router.ts	✔ Wired (added 2026-04-09)
dj	routers/dj.router.ts	✔ Wired (added 2026-04-09)
aiMix	routers/aiMix.router.ts	✔ Wired (added 2026-04-09)
projects	routers/index.ts (exported)	✔ Wired (added 2026-04-09)
presets	routers/index.ts (exported)	✔ Wired (added 2026-04-09)
settings	routers/index.ts (exported)	✔ Wired (added 2026-04-09)
ping	publicProcedure inline	✔ System health check

Key Procedure Contracts

ai.startPipeline

input: { sessionId: string; trackIds: string[] }
output: { pipelineId: string; nodeStatus: NodeStatus[] }

ai.submitSuggestionOutcome

input: { decisionId: string; outcome: 'accepted' | 'rejected' | 'ignored'; manualValueIfOverride?: number }
output: { logged: boolean; acceptanceRateUpdated: number }

sessions.endSession

input: { sessionId: string }
output: { metrics: SessionMetrics; exportPngUrl?: string }

14. Security & Compliance

Authentication Architecture

- Email/Password: bcrypt v6.0.0 → JWT (jsonwebtoken v9.0.3) → httpOnly cookie.
- Token Refresh: hydrateFromToken() race condition — FIXED.
- tRPC auth middleware: trpcAuth mount — FIXED and confirmed.

Authorization Layers

Layer	Method	Enforcement Point
Route-level	trpcAuth middleware	server/middleware/
Procedure-level	tRPC context validation	server/trpc.ts
Data-level	userId FK on every query	Drizzle ORM — all reads filtered
File access	Path traversal guard	Upload route — allowlist-validated
Subscription tier	Tier check in procedure	server/procedures.ts — all AI procedures

Rate Limiting

express-rate-limit v7.5.1 – 100 AI requests/minute/user
Message: 'AI request limit reached – upgrade to pro_artist for higher limits'

15. MVP Definition — Investor Checklist

Functional Requirements

Requirement	Status	Test
Load and play ≥8 simultaneous tracks	✔ Built	8-track test session
LLPTE node graph — animated, live inference	✔ Built	Node badge updates every 100ms
AI Auto-Leveling — active within 2 bars	✔ Built	Automated playback test
Ghost knobs visible on mixer strips	✔ Built	Visual QA
"Send to AI" context menu	✔ Built	Right-click on any clip
≥1 transition type (crossfade)	✔ Built	Execute and preview
Time Savings panel with real data	✔ Wired	SessionSummaryPanel in DAW.tsx
SessionChip in top nav	✔ Wired	Shows AI action count live
aiDecisionLog table	✔ Migration 0005	Schema + types + Zod schema
VU meters active on all 6 channel strips	✔ Built	Visual QA during playback
Effects rack: Reverb, Delay, Compressor	✔ Built	Knob interaction
Pro badge in top nav	✔ Built	Subscription tier check
TSC zero errors	✔ Verified	pnpm tsc --noEmit
Mix Suggestion System	❑ In Progress	MVP item 4 — frontend built, backend pending

Demo-Specific Requirements (PRD §26 — All Must Pass)

Requirement	Status	Failure Condition
inference 10ms badge updating live	✔	Static badge = demo failure
Transition Graph tooltip: edge count + tick time	✔	No tooltip = demo failure
Signal oscilloscope animated	✔	Static waveform = demo failure
Session summary exportable PNG	✔ Wired	Export fails = demo failure
aiDecisionLog writes live	❑	Acceptance rate shows 0 = weak demo
No audio dropouts in 20-minute demo	✔	Any dropout = demo failure
No crashes	✔	Any crash = demo failure

16. Risks & Mitigations

ID	Risk	Probability	Mitigation
R1	AI feels gimmicky	Medium	Ship only features with measurable, audible impact. Auto-Leveling first.
R2	LLPTE latency degrades under load	Medium	Confidence gating + auto-throttle. 8-track limit. Per-node tick monitoring live in UI.
R3	User distrust of AI decisions	Medium	Ghost knobs + confidence scores + easy override. Never auto-apply below 0.65.
R4	Demo crash in front of investor	Low	Pre-demo QA checklist — 17 items. Demo environment hardcoded pro_artist tier.
R5	Audio artifacts from AI adjustments	Low	All gain changes via AudioParam.setTargetAtTime() — click-free by design.
R6	Competitive response from Ableton/NI	Low	LLPTE is architectural — cannot be feature-flagged into legacy DAW.
R7	Inference cost exceeds revenue	Medium	Local inference for MVP — zero cloud cost. Post-MVP: quantized models on-device.
R8	Hygiene debt slows velocity	Medium	r3_hygiene.py patched: phantom dir exclusion, score formula uncapped, router key fixed (3 code fixes). 1 further fix flagged.

17. Post-MVP Roadmap

Phase 2 — Q3 2026 (6 months post-launch)

Theme: Depth + Community

Feature	Description	Rationale
Stem separation	AI-powered track separation from mixed audio	Unlocks mashup workflow for DJs
Custom AI preferences	User weights AI suggestions by genre/style	Reduces false positives for experienced users
Session history	Browse, replay, and fork past sessions	Retention — users revisit and improve past work
Hardware controller mapping	MIDI CC to mixer/effects parameters	Bridges R3 to existing DJ hardware
Collaboration (beta)	Real-time multi-user session sharing	Unlocks pro_artist tier use cases

Phase 3 — Q1 2027 (12 months post-launch)

Theme: Platform + B2B

Feature	Description	Rationale
LLPTE API (public)	tRPC endpoints exposed for third-party integrations	B2B revenue — DAW plugins, hardware OEM
Mobile companion	iOS/Android remote transport control	Extend studio to mobile without rebuilding core
Advanced analytics	Per-genre AI calibration, personalized baseline	Model improves with user data flywheel
Label/studio tier	Multi-user workspace with admin controls	Enterprise contract opportunity

Phase 4 — Q3 2027 (24 months post-launch)

Theme: AI Model + Market Expansion

Feature	Description	Rationale
R3 model fine-tuning	User fine-tunes LLPTE on own mix history	True personalization — highest retention driver
Generative transitions	AI generates entirely new transition content	Category-defining feature
Hardware partnerships	R3 software bundled with DJ controllers	Distribution at hardware scale
Marketplace	Community-shared AI presets and transition styles	Network effects — platform moat

18. Future Development & Upgrade Pathways

This section documents technical upgrade pathways not covered in the Phase roadmap. These are engineering-level decisions with long-horizon implications for the LLPTE architecture, infrastructure, and product surface.

18.1 LLPTE Model Upgrades

Current State

LLPTE uses heuristic confidence scoring — rules-based thresholds calibrated against 50-session alpha baselines. No trained model is deployed. aiDecisionLog (migration 0005) now captures labeled outcomes, beginning the data flywheel.

Upgrade Path

- Phase 1 (now → 100K decisions): Accumulate labeled data via aiDecisionLog.outcome field. Every accepted/rejected suggestion is a training signal.
- Phase 2 (100K decisions): Train first lightweight classification model on gain_adjust decisions. Deploy as WASM module in llpte-ai package — zero latency regression.
- Phase 3 (1M decisions): Per-user calibration. Rolling acceptance rate from mv_ai_acceptance_rates materialized view feeds confidence threshold adjustment per user.
- Phase 4 (post-Series A): Full per-genre model fine-tuning. LLPTE API exposed publicly — third parties can submit labeled audio decisions to improve the shared model.

18.2 Infrastructure Upgrades

Current

Railway (backend) + Vercel (frontend). Local inference only — zero cloud compute cost. PostgreSQL managed. S3/R2 for audio file storage (server/lib/storage-s3.ts via Proxy pattern).

Upgrade Path

- 10K users: Add mv_user_session_averages and mv_ai_acceptance_rates materialized views to PostgreSQL — currently missing, required for accurate Time Savings baseline calculation.
- 25K users: Migrate inference to dedicated GPU worker pool. AudioWorklet WASM modules remain local — only model inference moves to cloud. Estimated cost: \$0.0008 per inference at 25K users = ~\$600/month.
- 100K users: Read replicas for PostgreSQL. CDN for waveform peak data. OffscreenCanvas compute shaders for WebGPU waveform rendering already implemented — no change needed.

- Post-\$5M ARR: On-premise Docker deployment option (Dockerfile + docker-compose.yml already in repo). Required for enterprise/label tier SLA guarantees.

18.3 Audio Engine Upgrades

Current

Web Audio API + AudioWorklet (WASM module) + SharedArrayBuffer ring buffer. 8-track ceiling enforced at LLPTE layer.

Upgrade Path

- Track count expansion: LLPTE active edge limit is 2000 (current: 847). 16-track support achievable without architecture change. 32-track requires transitionGraph optimization pass.
- SIMD acceleration: Wasm SIMD already planned in architecture. Float32 SIMD for DSP operations. Target: 3ms processing per frame (currently undocumented baseline).
- WebGPU compute shaders: OffscreenCanvas already implemented for waveform rendering. Extend to FFT computation — moves spectralAnalyzer workload to GPU. Eliminates primary CPU bottleneck at high track counts.
- Stem separation (Phase 2): Demucs v4 quantized WASM model. Runs in AudioWorklet. 4-stem output (vocals, drums, bass, other). Estimated inference: 8–12 seconds per track on consumer GPU.

18.4 Frontend Architecture Upgrades

Current

React 18 + Vite 5 + Wouter + Zustand. DAW.tsx is ~1,800 lines — single-file monolith. Multiple phantom directories (client/client, client/hooks, client/stores) exist outside client/src/ and create import graph ambiguity.

Upgrade Path

- Immediate: Resolve phantom directory imports. client/src/store/auth-store.ts imports from ../hooks/authStore (phantom). Consolidate all stores under client/src/stores/, all hooks under client/src/hooks/.
- Near-term: Split DAW.tsx into feature modules. Suggested structure: DAW/TransportBar, DAW/ArrangementView, DAW/MixerSection, DAW/LLPTEGraph, DAW/EffectsRack. Each under client/src/features/daw/.
- Medium-term: React Server Components for sample browser and file tree. Reduces initial bundle. Vite 5 already supports RSC — no build tool change required.
- Long-term: Move to React 19 when stable. Concurrent features (useTransition, useDeferredValue) benefit the heavy FFT visualization renders without audio thread blocking.

18.5 Monetization Upgrades

Current

Stripe v20.4.1. Three tiers: explorer (free) / creator (\$20/month) / pro_artist (\$60/month). Subscription router wired. aiDecisionLog provides acceptance rate data required for upgrade prompt triggers.

Upgrade Path

- Contextual upgrade prompts: Wire aiDecisionLog acceptance rate to upgrade prompt logic. Trigger "upgrade to creator" when free user hits 5 AI uses in a session.
- Annual billing: Add annual_creator (\$192/year, 20% discount) and annual_pro_artist (\$576/year) Stripe price IDs. Server-side only — no schema change required.
- Usage-based billing (post-\$1M ARR): Charge per AI inference above tier limits. Stripe Meters API (introduced 2024). Requires inference counting middleware in sessionMetrics.service.ts.
- B2B API licensing (Phase 3): LLPTE tRPC endpoints exposed under /api/v1/llpte. Per-call pricing via Stripe Meters. Target customers: DAW plugin developers, hardware OEMs, content platforms.

18.6 Codebase Hygiene Upgrade Path

Current State (2026-04-09)

TSC errors	0
Routers wired	10/10
Live any violations	5 remaining: routes/presets.ts × 4 (Drizzle insert casts), audio-analysis.ts × 1 (eslint-disable exemption — no .d.ts for node-web-audio-api)
console.log violations	5 remaining in server/index.ts:300-308 (separate hard guard — replace with structured logger)
Phantom directories	9 remaining (client/client, client/hooks, client/components, client/stores, client/src/hook, client/src/context, client/src/contexts, client/src/store [LIVE — has imports], db/schema)
Hygiene script bugs fixed	3 code-fixed (phantom dir exclusion, score formula uncapped, router key subscription), 1 flagged (PRD items hardcoded — future fix)
Migration applied (Railway)	PENDING — 0005_overjoyed_gambit.sql

Path to 90/100 Hygiene Score

- P0: Apply migration 0005 to Railway production DB.
- P0: Wire aiDecisionLog writes into session-metrics.service.ts — enables live acceptance rate tracking.
- P1: Fix routes/presets.ts — 4 Drizzle as any casts — replace with typed InsertEffectPreset /

InsertEffectChain.

- P2: Replace console.log in server/index.ts:300-308 with structured logger (morgan already installed).
- P3: Consolidate phantom directories — merge client/client, client/hooks, client/components, client/stores into client/src/ canonical locations.
- P4: Create mv_user_session_averages and mv_ai_acceptance_rates materialized views in Drizzle migration 0006.
- P5: Fix vitest root config — add include pattern for packages/*/tests/*.test.ts to get accurate test count for PRD §0/§22/§25.

19. Go-to-Market Strategy

Channel 1: Content-Led (Primary — 0 to \$500K ARR)

- "Before AI / After AI" side-by-side audio comparisons (60 seconds)
- LLPTE node graph explanations — "This is what R3 actually does to your tracks"
- Time Savings reveal videos — "R3 just told me I mixed 42% faster"
- Distribution: Twitter/X, TikTok, YouTube Shorts, r/DJs, r/WeAreTheMusicMakers
- KPI: 100 signups per viral content piece at >100K views

Channel 2: Community (Primary — 0 to \$500K ARR)

- DJTT (DJ Tech Tools) — editorial relationship + sponsored content
- Discord DJ communities — authentic participation
- Creator economy newsletters (Matt Gielen, Paddy Galloway)
- KPI: 500 alpha users from community channels in first 90 days

Channel 3: Hardware Partnerships (Medium-term — \$500K to \$3M ARR)

Bundle R3 creator tier (3-month trial) with mid-tier DJ controller purchases via OEM agreement. Pioneer DJ, Rane, Denon DJ collectively ship 3M+ units annually. Target: 10% of 500K relevant units = 50K new users. At 10% conversion: 5,000 subscribers = \$100K MRR.

20. Financial Model & Valuation Gates

Cost Structure (Pre-Scale)

Infrastructure (Railway)	\$200/month
PostgreSQL (managed)	\$100/month
AI inference (local — MVP)	\$0 — no cloud inference cost
Storage (S3/R2)	\$50/month
Stripe fees	2.9% + \$0.30 per transaction
Error monitoring	\$26/month
Analytics	\$50/month
Total fixed/semi-fixed	~\$450/month

Valuation Gates

State	Valuation Range	Gap from Current
Current — technical asset only	\$180K–\$400K	Baseline
Working demo + 50 beta users	\$800K–\$2.5M pre-seed	4–6 weeks execution
≥65% AI acceptance rate confirmed	\$3–6M seed	Requires aiDecisionLog writes live
\$120K ARR (500 paying users)	\$4.8–9.6M (40–80× ARR, typical pre-seed SaaS)	12 months post-launch
\$1M ARR	\$15–30M Series A (15–30× ARR)	24 months post-launch

The current build state — LLPTE implemented, UI shipped, 11 routers wired, auth solid, database schema complete including aiDecisionLog — represents approximately \$300–600K of technical value as a pure asset. The gap between current state and seed metrics is approximately 4–6 weeks of focused execution.

21. Investor Demo Script (Updated — 12 Minutes)

Setup (30 minutes before)

- Run pre-demo QA checklist — 17 items confirmed (see §15)
- Load demo session: 8 pre-selected tracks, 128 BPM, key of A minor
- Confirm Pro badge visible in top nav
- Confirm LLPTE node graph visible and animated (all 5 connector lines moving)
- Confirm SessionChip in top nav is inactive (session not yet started)
- Dim room lights — screen is the only light source

Demo Flow

Minute	Action	Key Visual
0–1	Press Space. Say: "Watch the bottom-right corner."	LLPTE node graph animates. SessionChip shows ● Live.
1–2	Within 2 bars: toast fires "AI Mix updated — confidence 94%"	Ghost knobs appear on all channel strips simultaneously.
2–4	Click Accept All. Demonstrate one manual override.	Knobs animate to new positions. Override logged.
4–6	Suggestion panel fires: "Cut low-end on SYNTH LEAD below 80Hz"	"The AI caught a frequency clash in 6 seconds."
6–8	Right-click VOCAL CHOP → Send to AI → Key-Match Crossfade → Preview → Accept	Violet overlay. Camelot scoring. Transition committed.
8–10	Hover Transition Graph node — tooltip: "847 active edges last tick 0.8ms"	"10ms inference. Industry best is 15ms." Pause.
10–12	Press Space to stop. SessionSummaryPanel appears.	"42% faster. 18 minutes saved. 34 AI actions. 71% acceptance." Export PNG. Pause. "New category."

22. Glossary

LLPTE	Low-Latency Processing Transition Engine — R3's named real-time AI pipeline, 6 TypeScript monorepo packages
inputRouter	LLPTE node 1 — routes multi-track audio into pipeline, normalizes format
spectralAnalyzer	LLPTE node 2 — FFT, RMS, LUFS, true peak measurement per frame
aiMixEngine	LLPTE node 3 — inference engine, produces gain/EQ decisions with confidence scores
transitionGraph	LLPTE node 4 — precomputes and executes transition curves; 847 active edges in demo
outputBus	LLPTE node 5 — final signal delivery and monitoring
Ghost knob	UI element showing where AI wants to move a mixer parameter — translucent violet over actual knob
Confidence score	AI's self-assessed probability a decision is correct — gate at 0.65 for auto-apply
Confidence gating	Logic determining auto-apply (≥ 0.65), suggestion (≥ 0.40), or discard (< 0.40)
aiDecisionLog	PostgreSQL table logging every AI decision with outcome — the data flywheel. Migration 0005.
SessionChip	Top nav component showing live session state and AI action count. Wired 2026-04-09.
SessionSummaryPanel	Full-screen overlay on session end showing time savings, acceptance rate, export. Wired 2026-04-09.
Send to AI	Right-click context menu action routing audio clip into LLPTE pipeline for transition generation
Camelot wheel	Harmonic compatibility scoring for DJ mixing — used for key-match crossfade confidence
explorer	Free tier (correct DB value — NOT "free" or "Starter")
creator	Primary paid tier at \$20/month (NOT "Pro")
pro_artist	Studio tier at \$60/month (NOT "Studio")
Wire.txt protocol	Engineering discipline: Files Read → Findings → Changes → Remaining Ambiguities. No write without read first.
LUFS	Loudness Units Full Scale — broadcast-standard loudness measurement (ITU-R BS.1770-4)
Zinc-950	Tailwind CSS color token — darkest background in R3 design system (#09090b)
r3_hygiene.py	Python-based 10-phase monorepo audit script. 3 code bugs fixed 2026-04-09 (phantom dir exclusion, score formula, router key). 1 further fix flagged.

FINAL PRINCIPLE

This document is a contract, not a wish list.

Everything grounded in what is built, what is partially built, or what has a clear engineering path to completion.

**Prove that AI reduces the time and skill required to create a clean, professional mix
— in real time.**

The LLPTE pipeline makes that proof visible. The Time Savings panel makes it quantifiable.

The ghost knobs make it trustworthy. The 10ms inference badge makes it undeniable.

R3 v4 — PRD v4.0 — 2026 Q2